NOTE

Injector cannot be timed if the overspeed has been tripped. It must first be reset and the engine crankshaft barred over at least one revolution.

1. Bar engine over in the normal direction of rotation until flywheel pointer indicates the correct crankshaft position in degrees relative to top dead center of the cylinder being timed. Refer to setting instructions on Injector Timing Plate (located on right rear side of engine crankcase) and see Table 1 in Section 7 for top dead center settings.

2. Insert injector timing gauge into the hole provided for it in the injector body, Fig. 11-6.

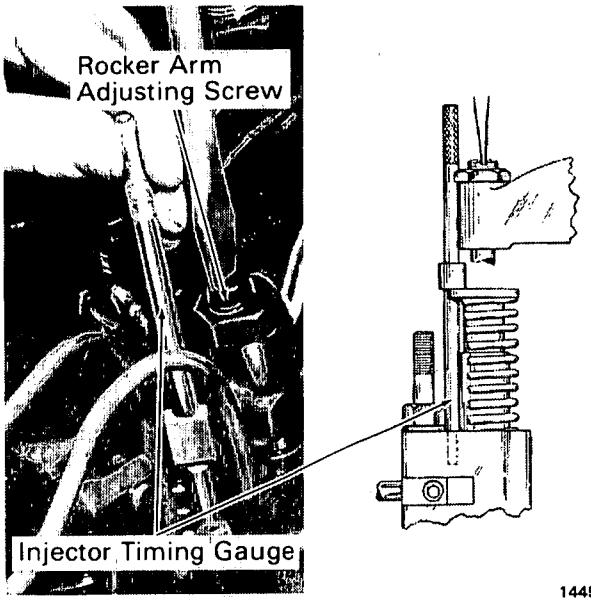


Fig.11-6 – Timing Injector

3. Loosen locknut and turn the rocker arm adjusting screw until the shoulder of the gauge just passes over the injector follower guide.

4. Tighten adjusting screw locknut while holding adjusting screw in position with screwdriver.

5. Recheck setting.

STICKING INJECTORS

Engines may encounter sticking injectors due to fuel, lube oil, or filter maintenance conditions. Since these conditions very often are momentary, injector removal may be minimized by utilizing alcohol to free up injectors while installed. This is done by applying ordinary commercial methanol to the injectors through a hole opposite the timing tool hole, and "popping" the injectors or motoring the engine. This sticking condition usually occurs on injectors which are held with the plungers down when the engine is stopped. Should injector racks show signs of sticking, they should be checked for gum or varnish deposits. If present, the rack should be cleaned with alcohol and rechecked. If sticking persists, the injectors should be removed and replaced with operational injectors. In no case should injectors be "crutched out" or cut out and the engine operated. If injectors operating unsatisfactorily cannot be remedied or replaced, the engine should be shut down until corrective action has been taken.

SERVICING INJECTORS

When servicing injectors, clean working conditions must be maintained. Dust or dirt in any form is a frequent cause of injector failure. When an injector is in an engine it is protected against dirt, dust, and other foreign materials by the various filters employed. When the injector is in storage, it is protected against harmful material by the filters sealing the body openings, which are in turn protected by shipping blocks.

However, an entirely different set of conditions is encountered when it becomes necessary to disassemble an injector for repair or overhaul. These conditions necessitate special shops, equipment, and trained personnel. It is recommended that non-operational injectors be returned to Electro-Motive for rebuild or unit exchange.

INJECTOR TESTER

In order to ensure efficient engine performance, injectors should be tested whenever removed from an engine, regardless of the reason for removal. It is advisable to test the complete engine set during each annual inspection. It is recommended that injectors be tested with the same oil used for protection against rust as given under "Storing Injectors".

It is important that the individual doing the testing understands the basic principles of injector operation and testing procedures in order to prevent acceptance of defective injectors and rejection of good ones. Instructions in the use of the injector tester and an outline of each separate test procedure along with a basic explanation of operation follows:

These injectors cover the testing of all needle valve injectors using the tester shown in Fig. 11-7. The procedures are not applicable to other types of testing equipment, since injector leakoff rates vary greatly in proportion to the volume of fuel contained in the high pressure portion of the tester.

HOLDING PRESSURE AND LEAK TEST

1. All injectors lose pressure due to leakage at any of several points, but this leakage must be controlled during injector manufacture to prevent engine lube oil dilution. The holding pressure test will qualify injectors having specified leakoff rates, providing this leakage is at the proper point and is satisfactorily controlled.

2. Manually hold the tester fuel line block on the injector. Pump until fuel is discharged from filter cap on opposite side, to remove air. Apply 12 411 kPa (1800 psi) to 13 790 kPa (2000 psi) pressure to the injector. No leakage is permitted at the nut to body seal, filter cap gasket, body plugs or between spray tip and injector nut.

3. Injectors should be qualified on the pressure test by timing the interval for a drop in pressure from 13 790 kPa (2000 psi) to 10 342 kPa (1500 psi). If this interval is less than 20 seconds (used) or 30 seconds (new or reconditioned), repeat the test, but close the pressure shutoff valve immediately after establishing the 13 790 kPa (2000 psi) pressure. This is to ensure that the leakdown time is not being affected by the possible leakage in the tester itself. If the timed interval for the pressure drop from 13 790 kPa (2000 psi) to 10 342 kPa (1500 psi) is still less than 20 seconds (used) or 30 seconds (new or reconditioned), the injector should be rejected. To relieve the pressure before removing the injector from the tester, wrap a cloth around the injector fuel line connections and back off on the clamping wrench, Fig. 11-7.

RACK FREENESS TEST

1. The rack engages with a small pinion on the injector plunger and serves to rotate the plunger with respect to two ports in the injector bushing, which regulates the amount of fuel injected with each stroke of the plunger. Binding of the rack is generally caused by damaged gear teeth, scored plunger and bushing, or galling of rack itself. A binding rack may cause sluggish or erratic speed changes and overspeed trip action.

2. To be considered satisfactory, the rack must fall in and out through full travel by its own weight when injector is held horizontally and rotated about its axis.

BINDING PLUNGER TEST

1. Failure of the injector plunger to move up and down freely indicates scoring of the plunger and bushing or weak or broken spring. A binding plunger will cause erratic cylinder firing and, in extreme cases, overspeed trip action.

2. Place injector in tester but do not attach the fuel line. Place rack in the full fuel position and pump all the fuel out of the injector with injector popping lever, Fig. 11-7. When all of the fuel has been removed, depress the injector plunger to full extent of its travel. Slowly release popping lever and simultaneously move injector rack repeatedly in and out through its full travel.

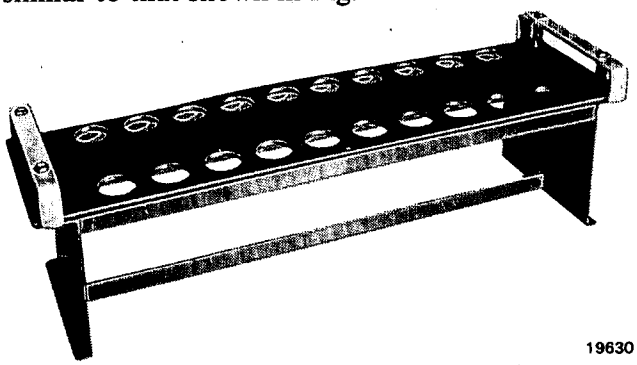
REPLACING INJECTOR FILTERS

Injector filters should not be disturbed or removed except during injector reconditioning (when all parts are completely washed), or in the event of fuel stoppage to the injector.

STORING INJECTORS

When injectors are not to be used for a considerable length of time, they should be protected against rust by using a stable, noncorrosive straight-run petroleum distillate in the kerosene volatility range. It is also recommended that injectors be tested using this oil. If this is done, treatment will be taken care of at time of injector test.

After treatment, the injectors should be stored in a protective container until needed. This container should accommodate an injector holding rack similar to that shown in Fig. 11-8.



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Fig. 11-8 – Injector Holding Rack

INJECTOR LINKAGE

DESCRIPTION

The injector linkage, Fig. 11-9, consists of the mechanical arrangement between the governor and the injector permitting all injector rack positions to be changed simultaneously when the governor terminal shaft is rotated. Two injector control rods connect the lever on the governor terminal shaft to